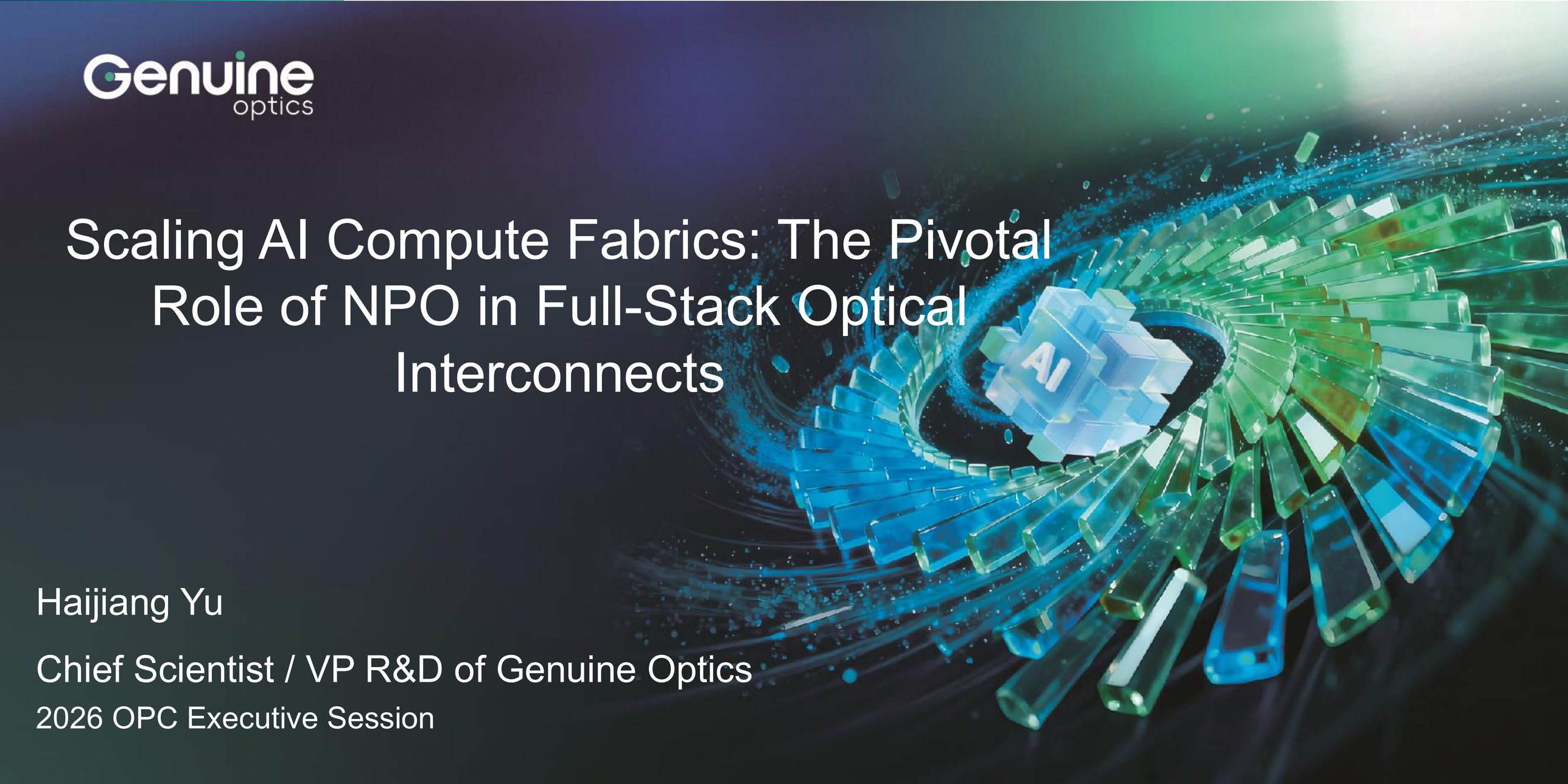
Scaling AI Compute Fabrics: The Pivotal Role of NPO in Full-Stack Optical Interconnects



Haijiang Yu

Chief Scientist / VP R&D of Genuine Optics

2026 OPC Executive Session

Who Are We - Company At a glance

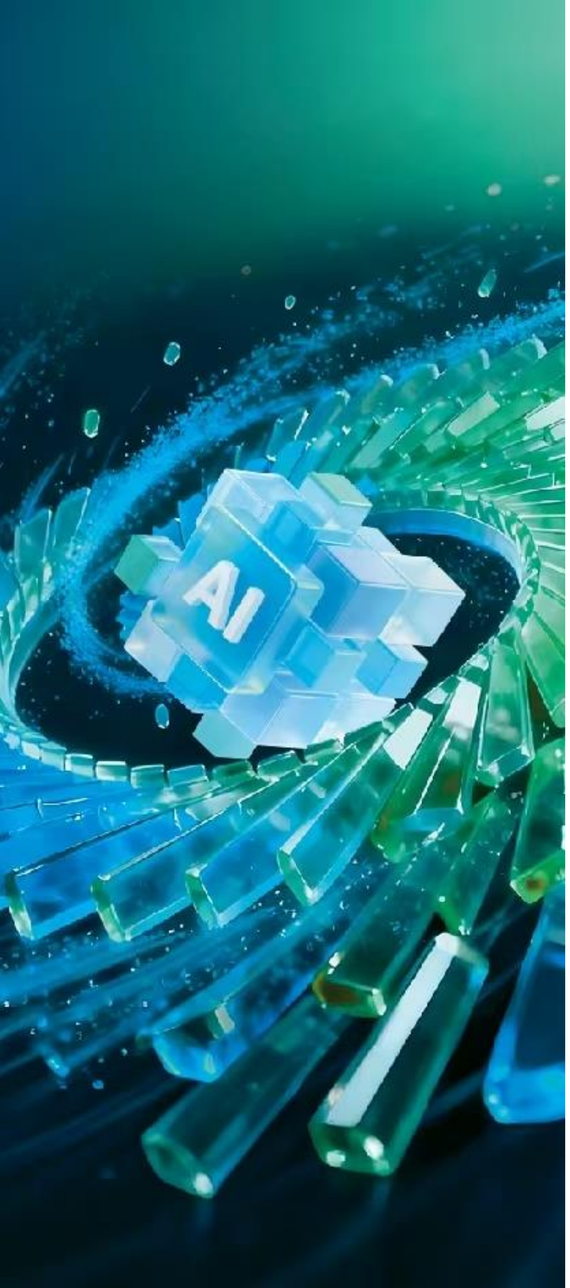
- Genuine Optics serves **global customers** across **multiple** continents today.
- A **Top 8 optical module** supplier in 2024.
- Globally integrated teams deliver **world-class** execution and support.
- Delivering **>millions of Transceivers** to meet your needs.



Genuine optics
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Corporate Headquarters

Genuine optics
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Chonburi, Thailand

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Contents

Market Driver: Exponential growth of LLMs demanding high-throughput, low-energy, low latency optical fabrics.

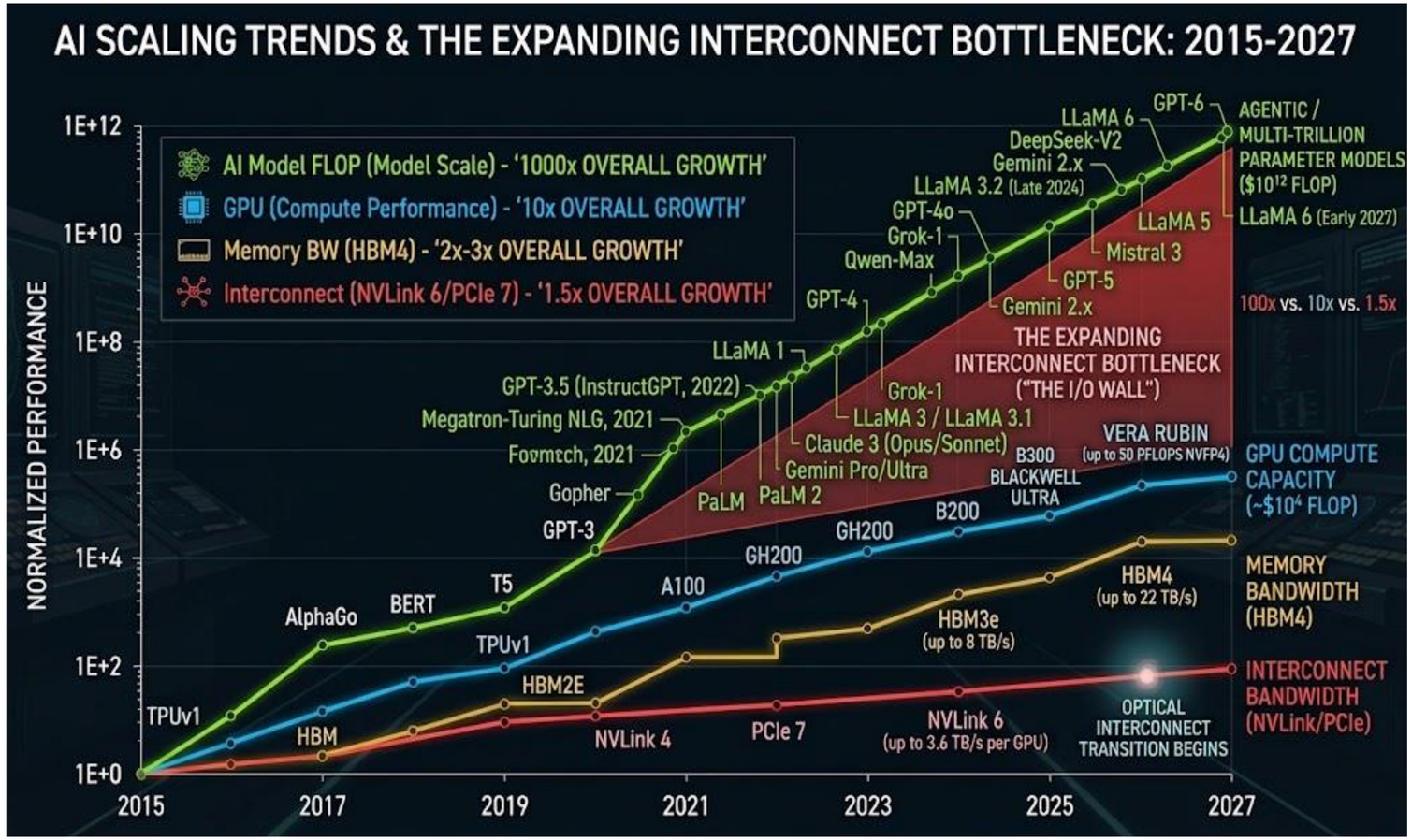
Full-Scale roadmap: Form-factor migration (Pluggable →NPO →CPO) and per-lane speed scaling (100G → 200G → 400G).

Foundational Tech: In-house SiPh/TFLN platform + high-power\high-efficiency External Light Sources (ELSFP/ELS).

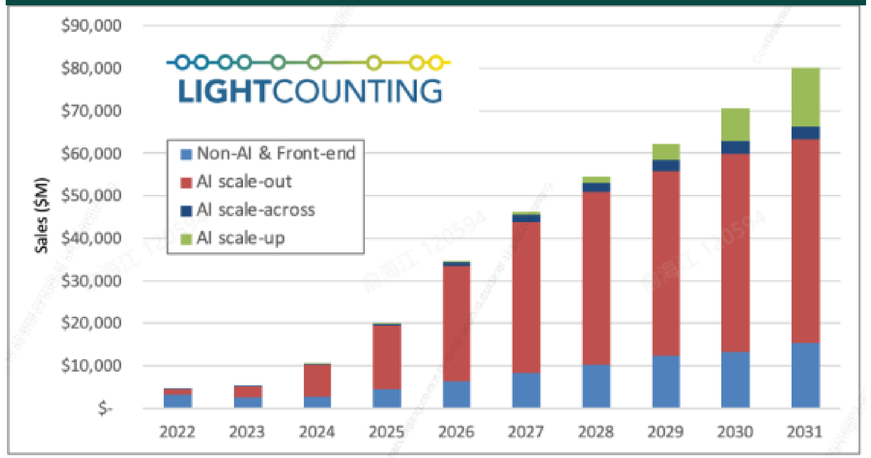
Conclusion

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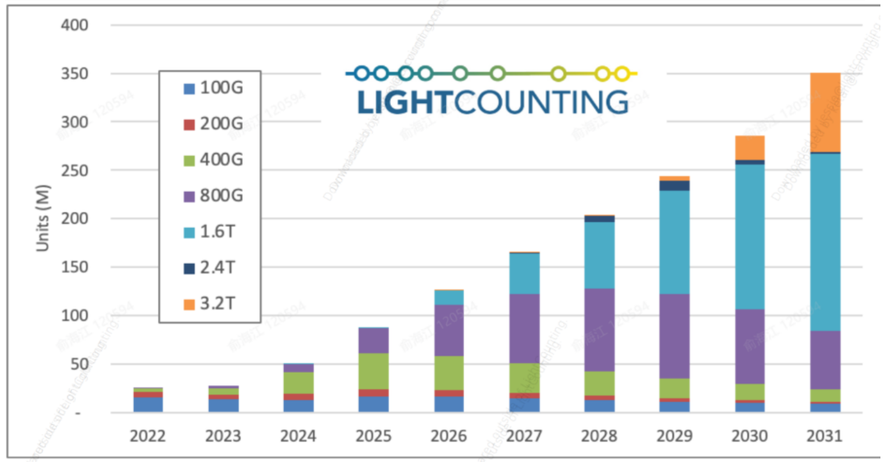
Soaring AI Compute: Driving Exponential Growth in Optical connectivity



Shipments of pluggable and NPO/CPO by application



Shipments of pluggable and NPO/CPO by speed



Source: LightCounting

Biennial Growth Rate of AI Computing Center Infrastructure

- AI Training Computing Power: ~100X
- HPC/GPU: 3.3X
- Memory BW = 1.4X
- Interconnect = 1.4X

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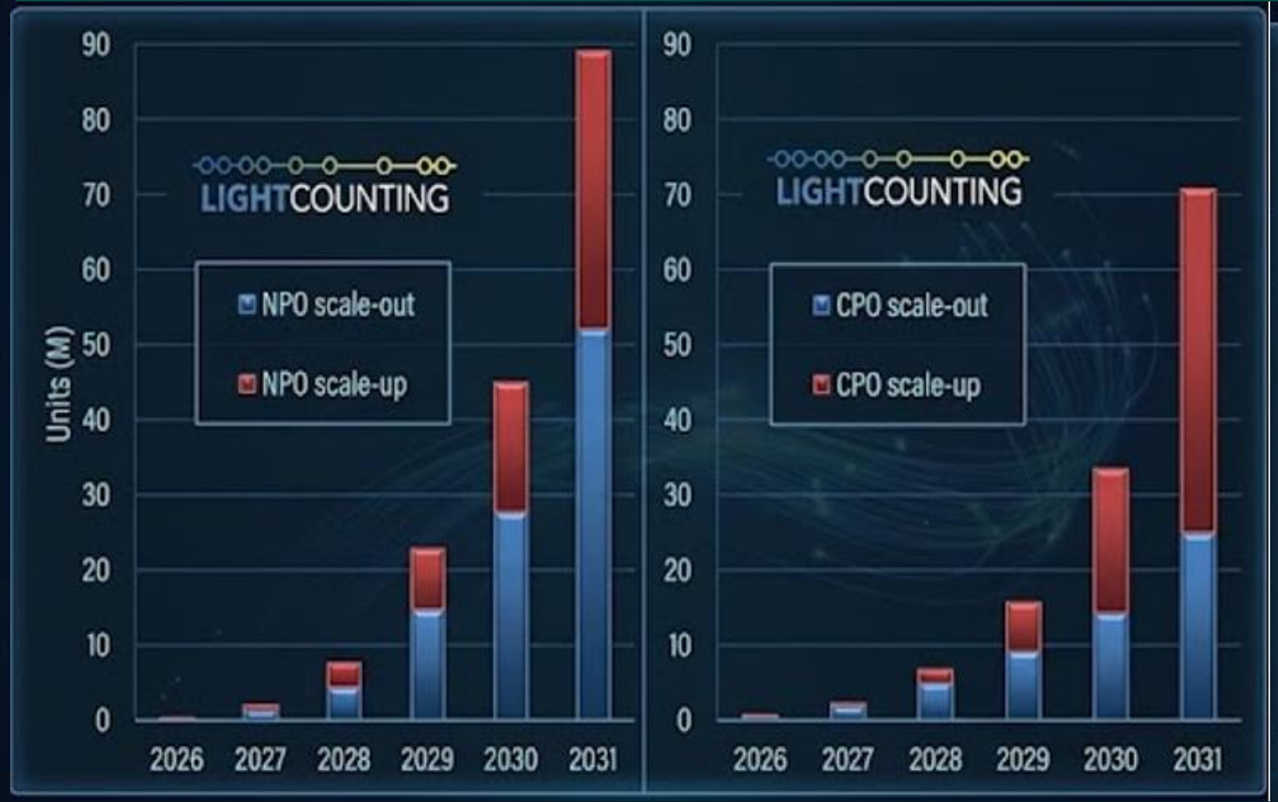
Next-Gen AI Optical Fabrics: Pluggable vs NPO/CPO Growth Trends

Genuine optics

Shipments of pluggable and NPO/CPO ports



Shipments of NPO/CPO by application



Market Transition: Retimed pluggable modules dominate short-term deployments, while LPO, NPO, and CPO experience exponential growth starting around 2027–2028 to address power and BW density bottlenecks.

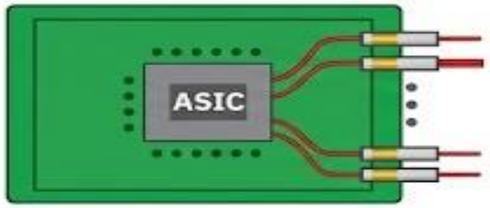
Architecture Divergence: NPO adoption is primarily led by scale-out network demands, whereas CPO volume is heavily driven by scale-up GPU/accelerator fabrics and volume deployment is expected to lag behind NPO .

Evolution of Optical interconnect technologies

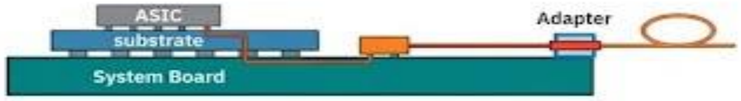
Genuine optics

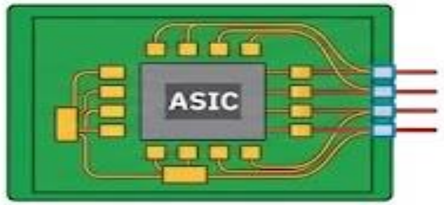
Pluggable Optics





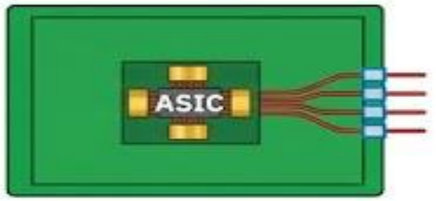
NPO (Near-Packaged)





CPO (Co-Packaged)





Feature / Dimension	Pluggable Optics	NPO (Near-Packaged)	CPO (Co-Packaged)
Optics Position	Front panel edge	Host board (<10mm to ASIC)	Directly on ASIC package
Power Efficiency	Baseline High loss	+30-50% Savings	+50%+ Savings
Thermal Management	Easy Independent airflow	Manageable Separate cooling	Complex High thermal density
Field Repairability	100% Hot-Swap	High Modular sockets	Extremely Difficult Board scrap
Ecosystem Readiness	Mature Multi-vendor	Emerging Standard (OIF)	Early R&D / Proprietary

NPO is a pragmatic solution among performance, engineering feasibility and ecosystem readiness

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GO Scalable Optical Connectivity Roadmap for AI Computing networks

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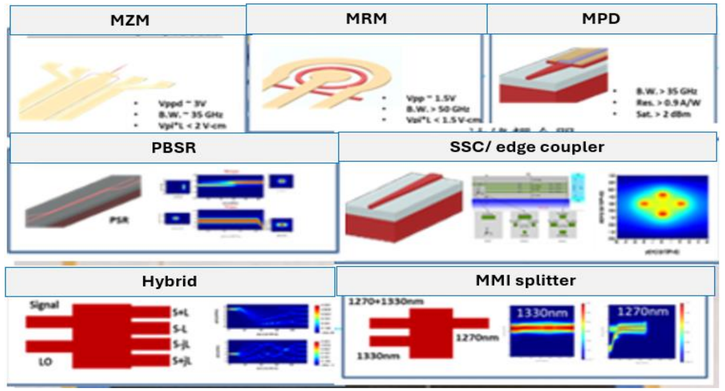
Aggregate BW: 800G → 1.6T →3.2T →6.4T → 12.8T
Per-lane Speed: 100G → 200G → 400G
Connectivity solution: TRO/LRO/LPO → XPO/NPO/CPO

	2023	2024	2025	2026	2027	2028+
Pluggable Modules	400G DR4	800G DR8	1.6T DR8	12.8T XPO / 1.6T DR4	3.2T DR8	25.6T XPO?
NPO		3.2T TRx MZM PIC	3.2T NPO OE / ELSFP	6.4T NPO	12.8T NPO	
CPO/OIO		3.2T TRx MRM PIC	3.2T MRM CPO OE / DWDM ELSFP		400G/800G OCI OE	1.6T+ OCI OE

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End-to-End In-house SiPh/TFLN platform

Design Platform



- **PIC design automation:** Automate/AI assisted simulation & design work flows
- **Proprietary PDK library:** Custom device building blocks
- **System-level Co-simulation:** Complete end to end link optimization

Foundry Platform



- **Proven Tape-Out Track Record**: >30 MPW & full-reticle tape-outs across SiPh/TFLN
- **Multi-Foundry Supply Chain**: >3 Tier-1 global foundries outside China
- **High-Speed Scaling**:
  - 100G & 200G/lane: Mass production
  - 400G/lane: module live demo
  - 3.2T NPO/CPO OE: Live demo

Product Platform





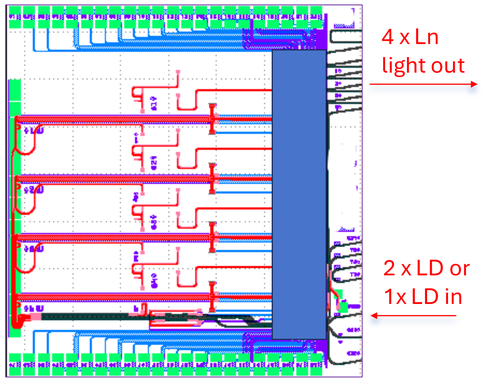
- **Full-Stack Automated Testing**: Die, wafer, and module-level characterization
- **100% known good die (KGD):** Rigorous wafer-level screening
- **Comprehensive Qualification**: EVT, DVT, PVT
- Telcordia-Grade Reliability: High-volume burn-in & life testing

Our in-house end-to-end vertical integration delivers Next-Gen optical connectivity with silicon photonics solutions with speed, reliability and proven scale.

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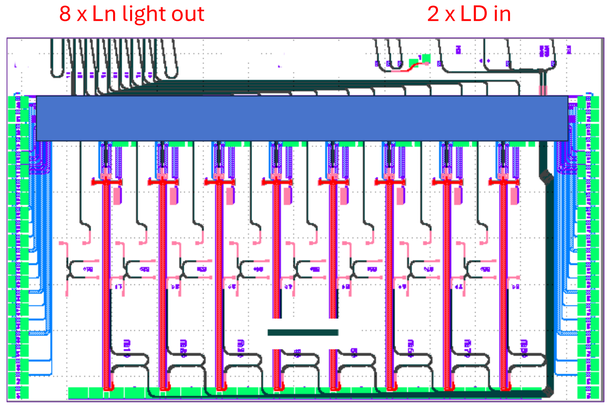
Key SiPh PIC products

DR4 Tx -400G/800G/1.6T



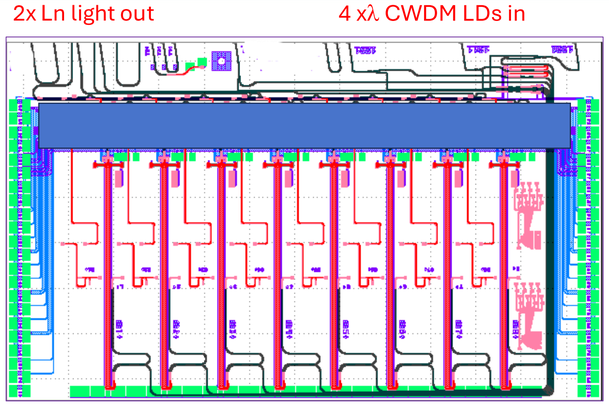
- **BW:** > 50GHz
- **Vpi**: <6V
- **Coupling loss:** < 1.5dB/facet
- **Laser sharing:** 1x2 or 1x4
- **Heater Pπ:** <15mW
- **DFM/DFR:** WLT / OCIl, PD, SSC

DR8 Tx- 800G/1.6T



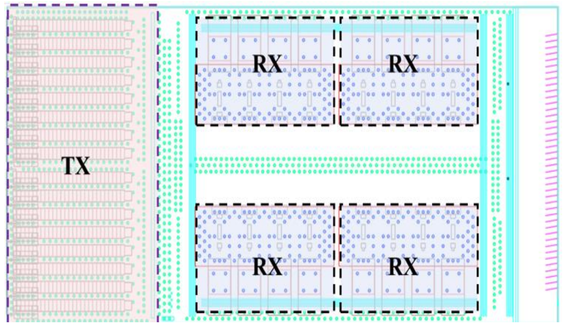
- **BW:** 50~60G GHz
- **Vpi**: <6V
- **Coupling loss:** < 1.5dB/facet
- **Laser sharing:** 1x4 or 1x8
- **Heater Pπ:** <15mW
- **DFM/DFR:** WLT / OCIl, PD, SSC

2FR4 Tx -800G/1.6T



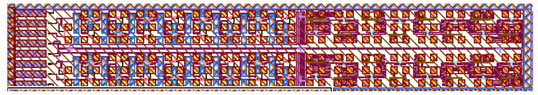
- **BW:** > 50GHz
- **IL , Vpi**: <6dB/ch, <6V
- **Coupling loss:** < 1.5dB/facet
- **Laser sharing:** 1x2
- **Heater Pπ:** <15mW
- **DFM/DFR:** WLT / OCIl, PD, SSC
- **MUX**: Compact Brag Grating, BW: 20nm, X-talk: **-25dB**

3.2T/6.4T MZM TRx For NPO



- **32-CH MZM/WG-PD TRx**
- **100G or 200G/lane**

3.2T MRM TRX for CPO



- **2 × 8λ 32-CH MZM/WG-PD TRx**
- **100G or 200G/lane**

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SiPh PIC: Field Proven Reliability

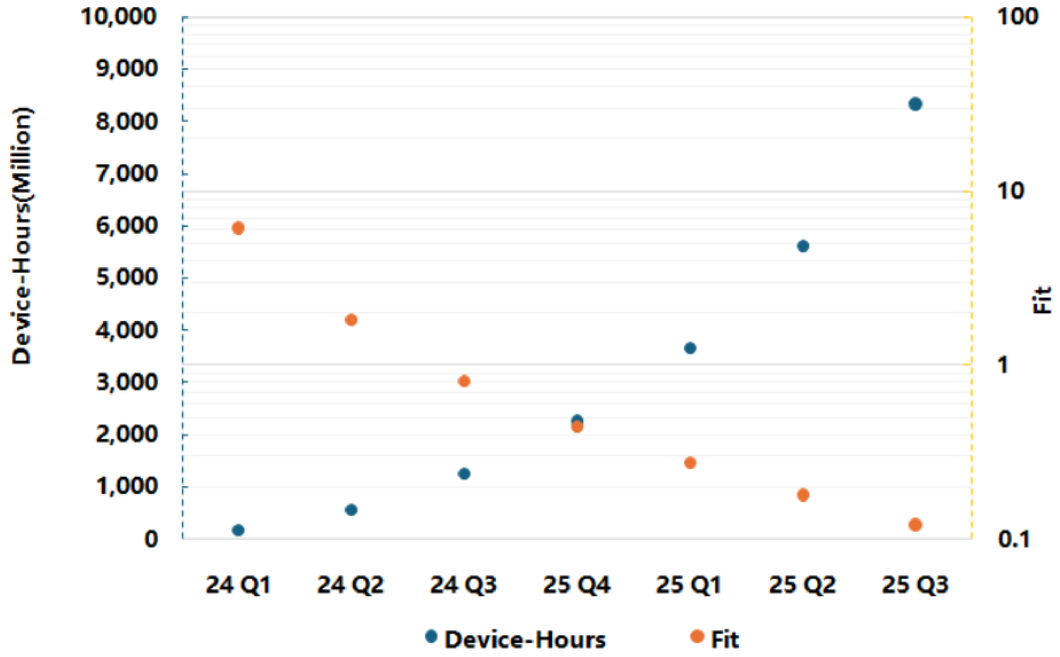
Million-Scale Deployment: Over a million silicon photonic modules shipped.

Proven Ultra-Reliability: >8 billion device hours with a PIC FIT rate of **<0.2**.

$$\lambda = \frac{\gamma}{T_{\text{total}}} \times 10^9$$

where:

- γ is the number of random failures.
- T_{total} is the total working time of all devices in field deployment.



*Note: The above data is **based on 2 -year shipment**. leadership in the industry. With the continuous shipment of GO silicon photonics products, the FIT rate will be further reduced.*

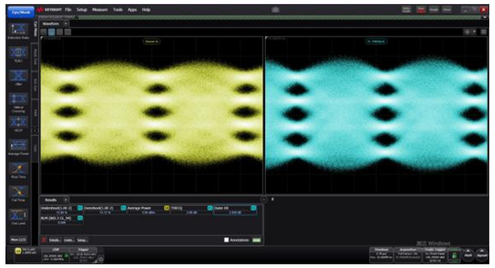
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In-house 200G/lane SiPh solutions for 1.6T OSFP products

1.6T Module Key Features

Parameters	Full DSP	LRO	LPO
Power consumption	< 24W	< 15W	< 10W
Loopback BER	~E-13 Level	~E-13 Level	< E-8 Level

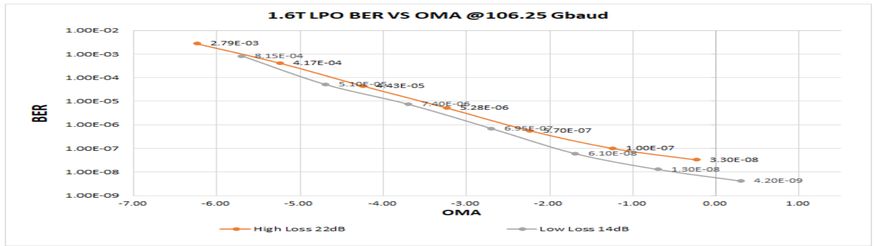
1.6T LPO TP2/TP4 eye and BER



TP2 Eye: TDECQ = 2.25dB

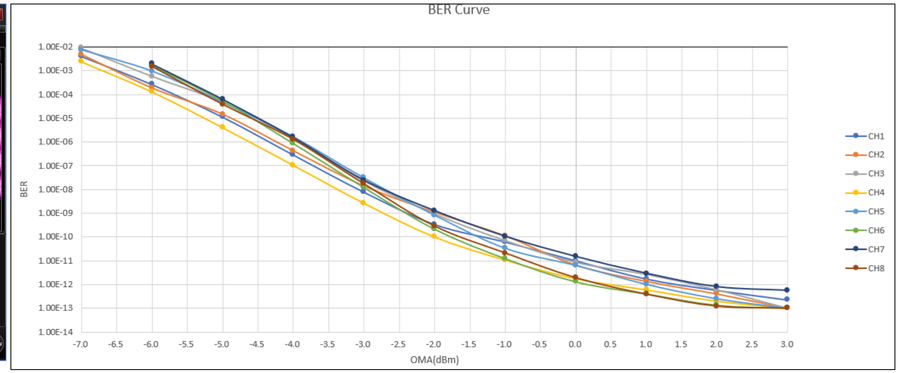


TP4 Eye: TDECQ = 3.11dB



1.6T OSFP DR8(Full DSP) TP2 Eye and BER (test board results)

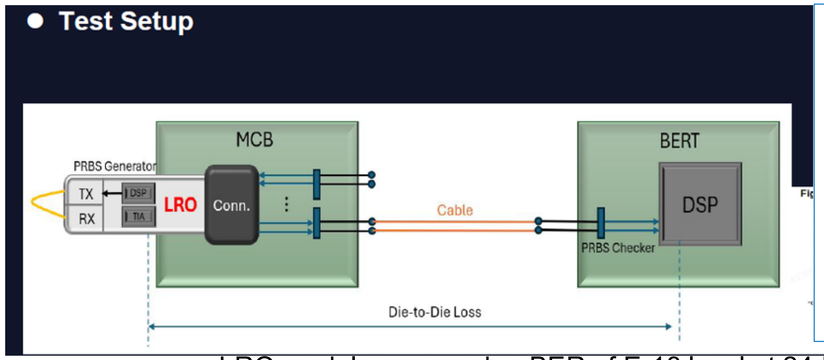


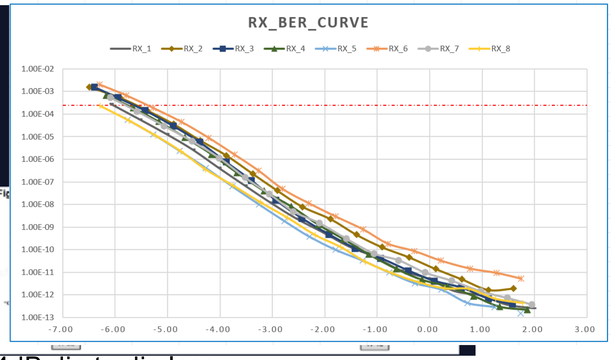


- TDECQ: 1.65dB ER: 4.2dB
- RLM: 0.986

- DPO Module can reach a BER down to E-13

1.6T LRO BER Test Data (test board results)



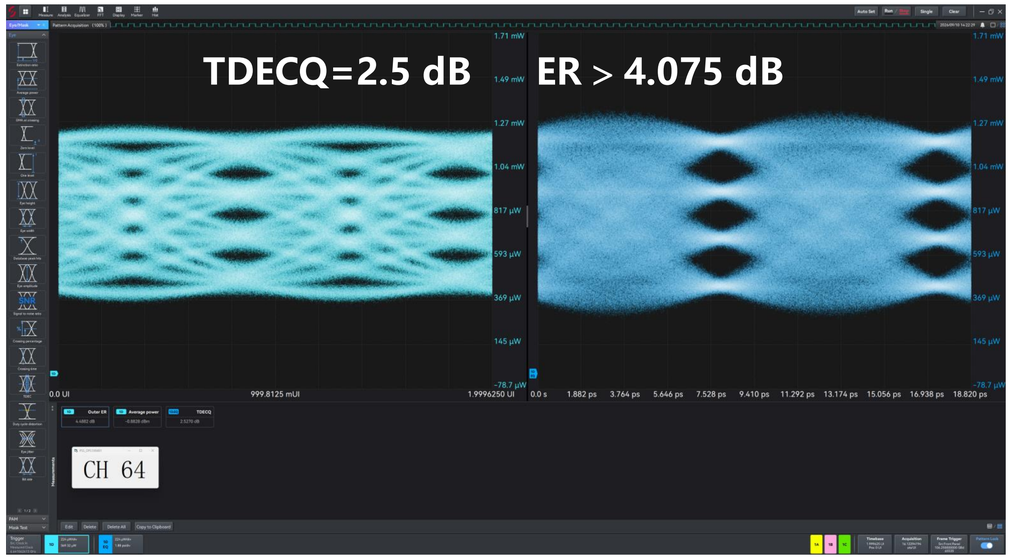


- LRO module can reach a BER of E-13 level at 24dB die to die loss

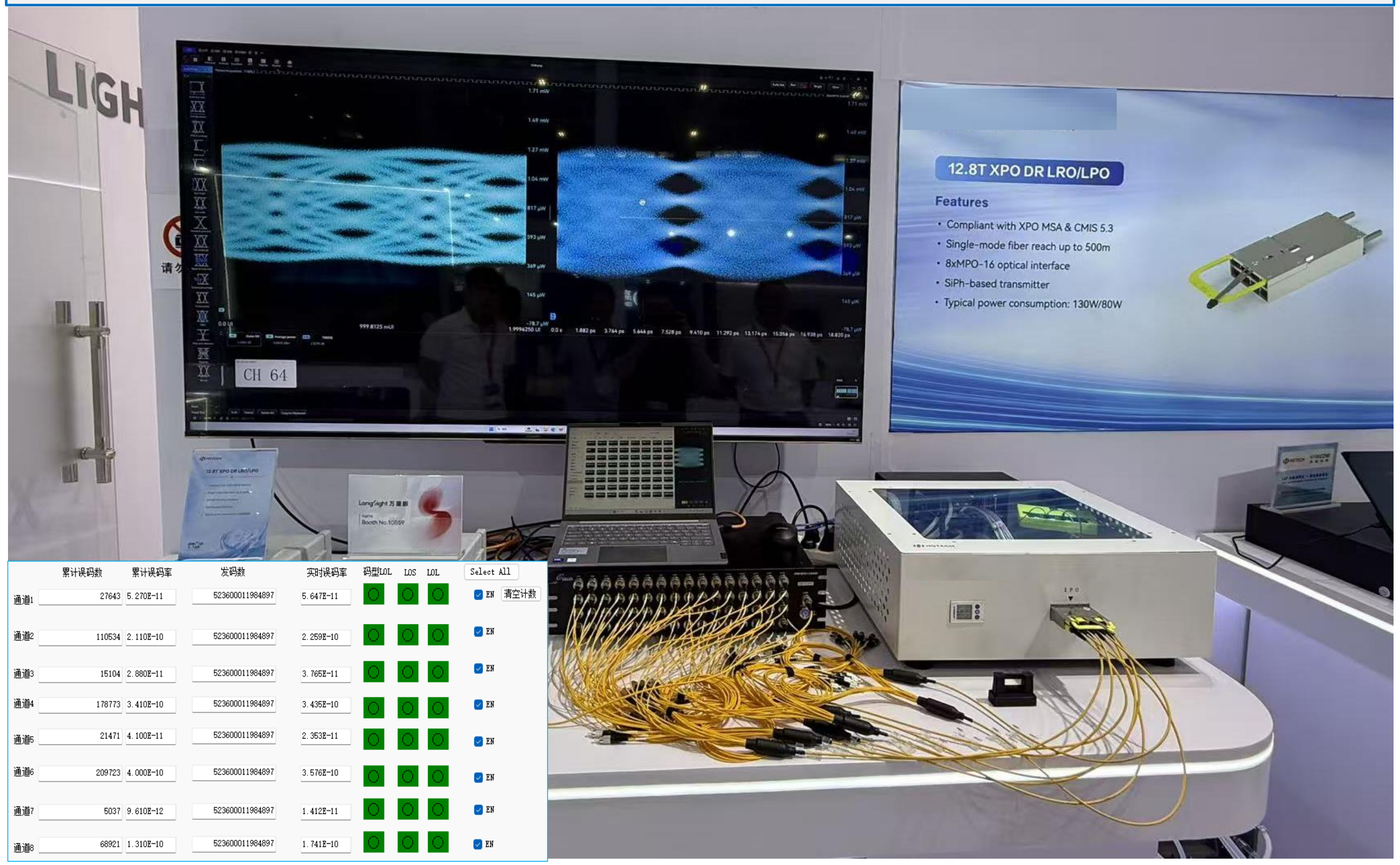
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12.8T XPO: Pluggable solution for Scale-up and Scale out

- Live Demo in Booth E4



GO 12.8T XPO live Demo in 2026 Shenzhen CIOE



Features:

- Compliant with XPO MSA & CMIS 5.3·
- Single-mode fiber reach up to 500m
- 8xMPO-16 optical interface
- In-House SiPh-based transmitter
- Typical power consumption: 130W/80W

Performance::

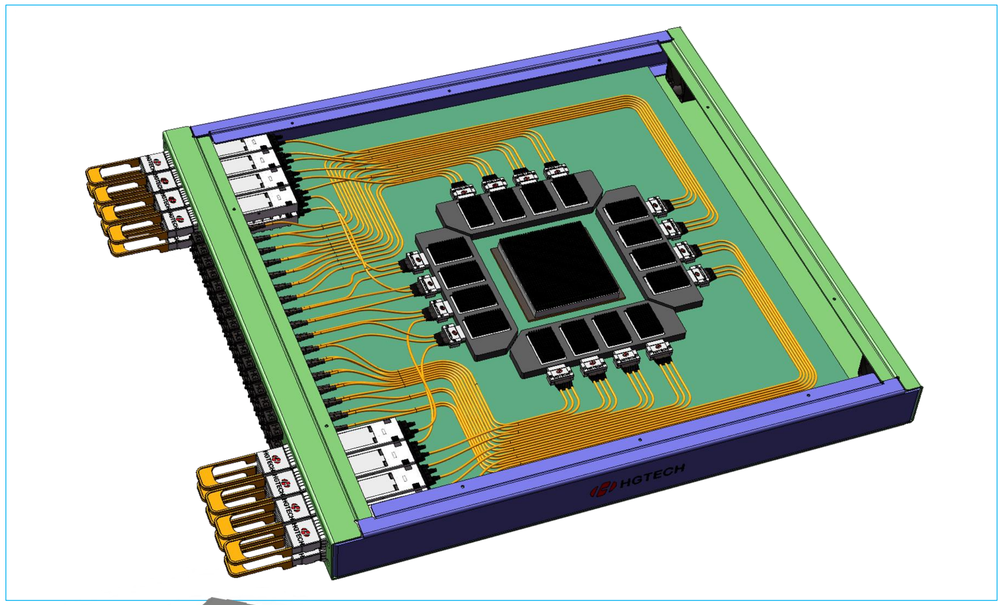
- LRO (DSP Sian 3-L)
- Power consumption: < 130W
- TDECQ：2.5dB
- ER： > 4dB
- BER：E-10~E-11

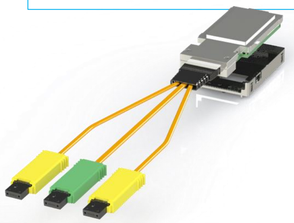
3.2T / 6.4T NPO Solutions (100G/200G per-lane SiPh)



ELSFP

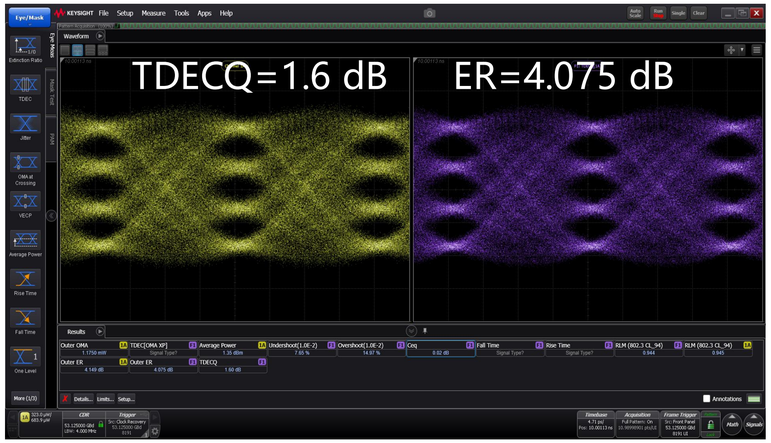
- Power consumption < 8.5 W
- Output optical power ≥ 20 dBm/lane
- 12-fiber MT interface, 8 PMF
- Central wavelength: 1311 nm
- Operating temperature: 0–70 °C
- Compliant with OIF specifications

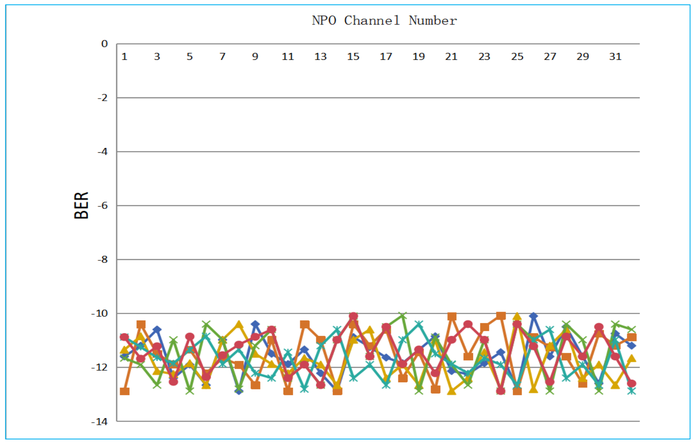




3.2T / 6.4T Optical Engine

- Max. 32 channels, 100/200 Gbps/lane
- Highly integrated PIC （SiPh）
- Highly integrated FAU
- Low power consumption
- Compliant with OIF specifications
- Operating temperature: 0–85 °C

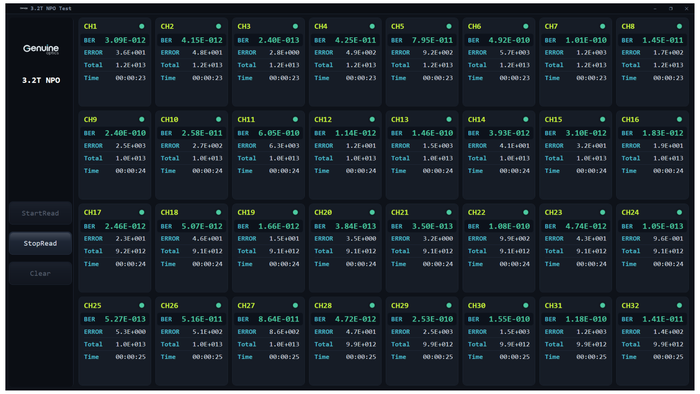




Loopback BER on EVB < 1E-10

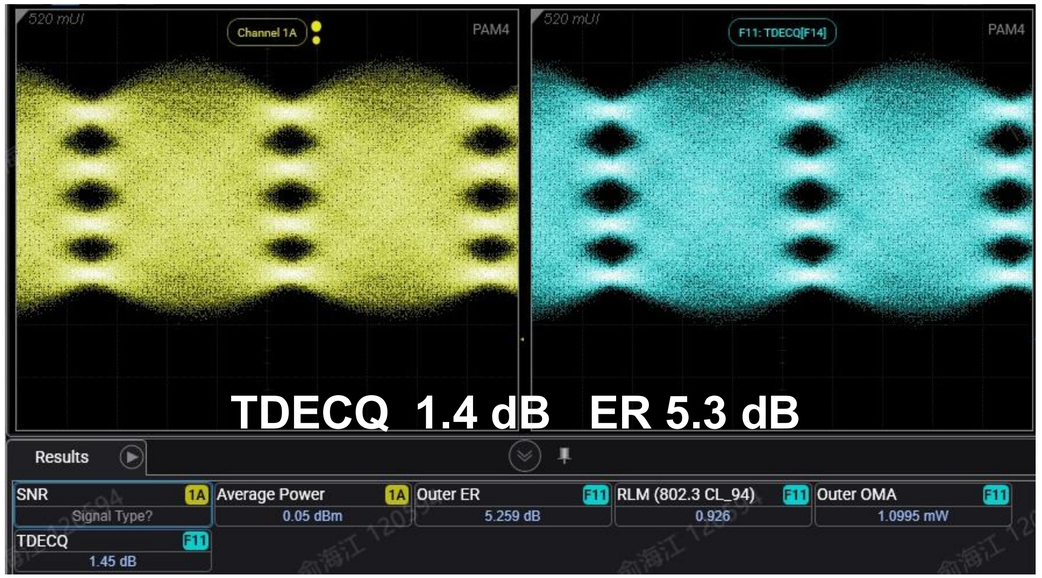
GO 3.2T NPO live Demo in 2026 Shenzhen CIOE





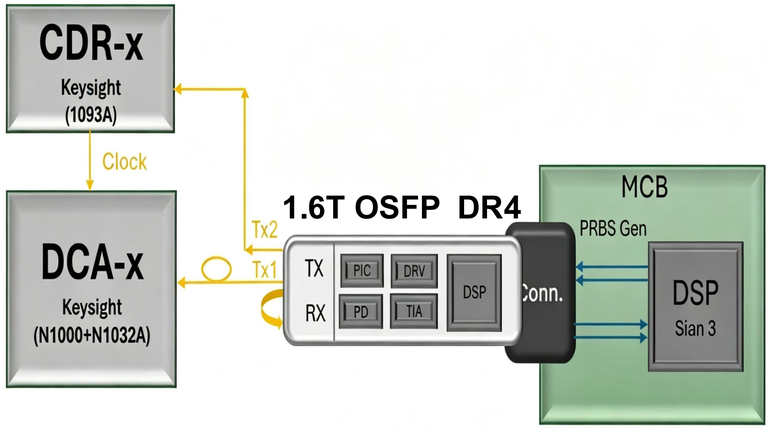
- 6.4T NPO demo in booth E4
- Scalable to 12.8T NPO by using GO’s 400G/lane technology

400G/lane module solution for 3.2T

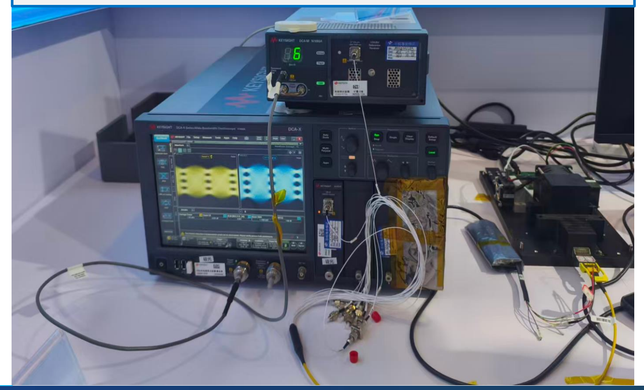


BER: ~E-6

```
start check Media PRBS_MON status
.
PRBS lock = 0x1,  loss_of_lock_sticky = 0x0, err_cnt = 0x25682cf ber=8.014316e-006
.
PRBS lock = 0x1,  loss_of_lock_sticky = 0x0, err_cnt = 0x54767fe ber=8.021735e-006
.
PRBS lock = 0x1,  loss_of_lock_sticky = 0x0, err_cnt = 0x8459a5d ber=7.976092e-006
.
PRBS lock = 0x1,  loss_of_lock_sticky = 0x0, err_cnt = 0xb20dcd5 ber=8.025981e-006
.
PRBS lock = 0x1,  loss_of_lock_sticky = 0x0, err_cnt = 0xe1dac45 ber=8.080054e-006
.
PRBS lock = 0x1,  loss_of_lock_sticky = 0x0, err_cnt = 0x1100ec61 ber=8.079991e-006
.
PRBS lock = 0x1,  loss_of_lock_sticky = 0x0, err_cnt = 0x13f2b1c3 ber=8.043993e-006
.
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.
PRBS lock = 0x1,  loss_of_lock_sticky = 0x0, err_cnt = 0x19c87ede ber=8.084610e-006
.
PRBS lock = 0x1,  loss_of_lock_sticky = 0x0, err_cnt = 0x1cab6f09 ber=8.047184e-006
```



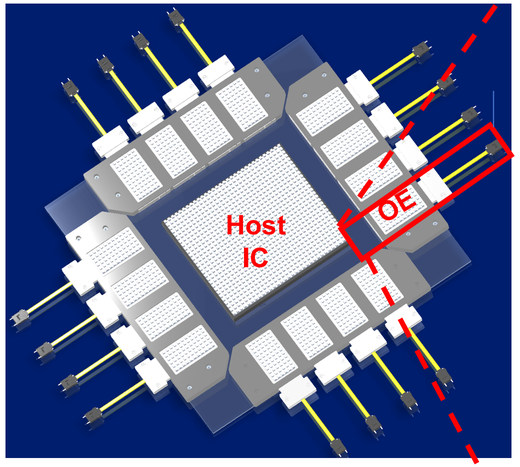
GO 1.6T DR4 live Demo in 2026
Shenzhen CIOE



Features

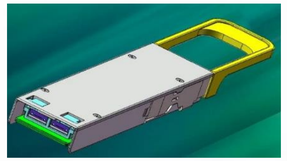
- Electrical: 8 × 212.5 Gbps PAM-4
- Optical: 4 × 425Gbps PAM-4
- In-house SiPh/TFLN + 3nm 400G DSP
- 100mW CW laser with 1x4 split
- Operation temperature 0-70C
- Power consumption: ~30W
- 500m reach
- Live demo in booth E4
-

3.2T/6.4T MRM CPO OE and DWDM ELSFP – OCI-MSA solution

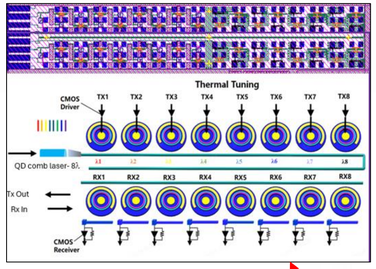


Concept drawing of CPO integrated on glass substrate

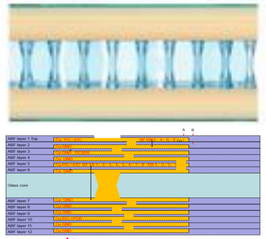
QD 8λ DWDM Comb ELSFP

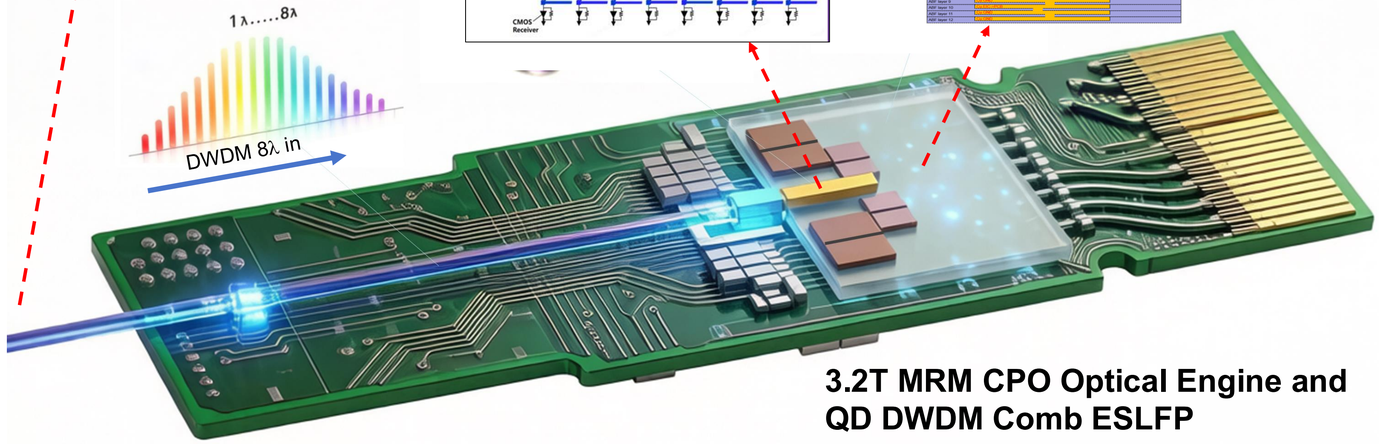


2×8λ ×200Gbps 3.2T MRM TRx



12- layer TGV glass substrate

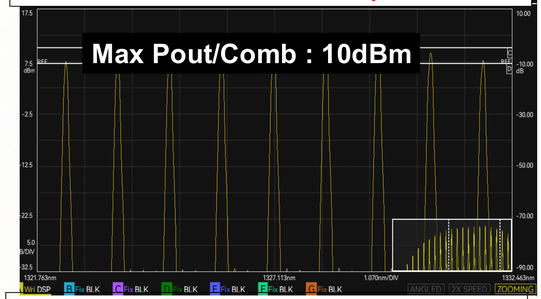




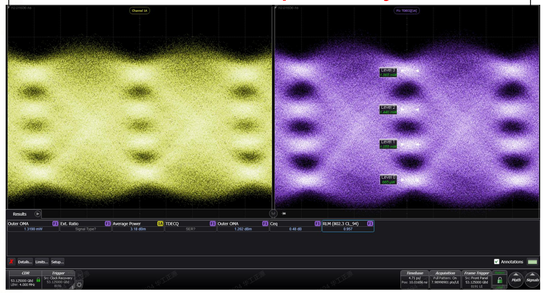
3.2T MRM CPO Optical Engine and QD DWDM Comb ESLFP

- DWDM ELSFP live demo in booth #1134

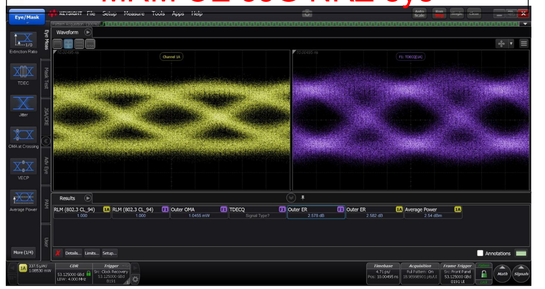
QD Comb laser ELS spectrum



MRM OE optical eye



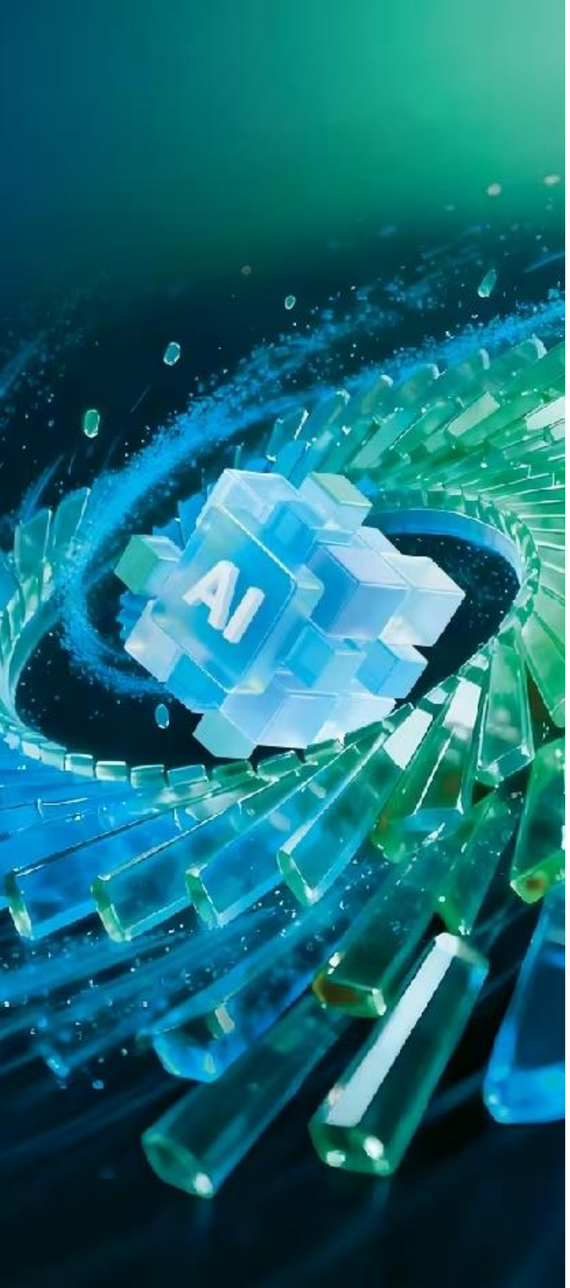
MRM OE 53G NRZ eye



2 × 8λ × 200 Gbps 3.2T MRM TRx
Quantum dot 8λ DWDM comb laser (ELS)
TSV/TMB-based 2.5D/3D packaging
Bandwidth density > 3 Tbps/mm (unidirectional)

- Scalable micro-ring CPO OE+, DWDM ELSFP and TGV glass substrate, with live-demo in OFC 2026/CIOE 2026
- Paves the way for OCI-MSA solution

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Conclusion

The AI Imperative: Full-stack optical connectivity is critical to breaking power, latency, and throughput bottlenecks in next-gen AI compute clusters.

Foundational Core: In-house Silicon Photonics (SiPh), Thin-Film Lithium Niobate (TFLN), and External Light Source (ELS) platforms form the indispensable baseline.

Scalable Architecture: Continuous SiPh innovation across 800G to 12.8T delivers scalable density and power efficiency to meet the next-gen AI network demand.

Pragmatic Evolution: Driving parallel paths across XPO, NPO, and CPO—NPO represents the optimal sweet spot for immediate commercial adoption, paving the pragmatic path toward ultimate CPO deployment

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400G/Lane for 3.2T

12.8T XPO

6.4T NPO